



AIR FORCE COMMUNITY PROGRAM

Military Resource Exchange (MRE)

Air Force Community Equipment Exchange & Accountability Node. The MRE is a centralized logistics hub designed to streamline PCS transitions — managing Class II through IX resources with community accountability and automated intelligence.

Built on Laravel 11

Modern Stack

Built with **Laravel 11** and **PHP 8.3**, leveraging the latest performance improvements and security patches in the PHP ecosystem.

MVC Architecture

The Model-View-Controller pattern enforces clean separation of concerns — keeping business logic, data, and presentation layers distinct and maintainable.

Local Deployment

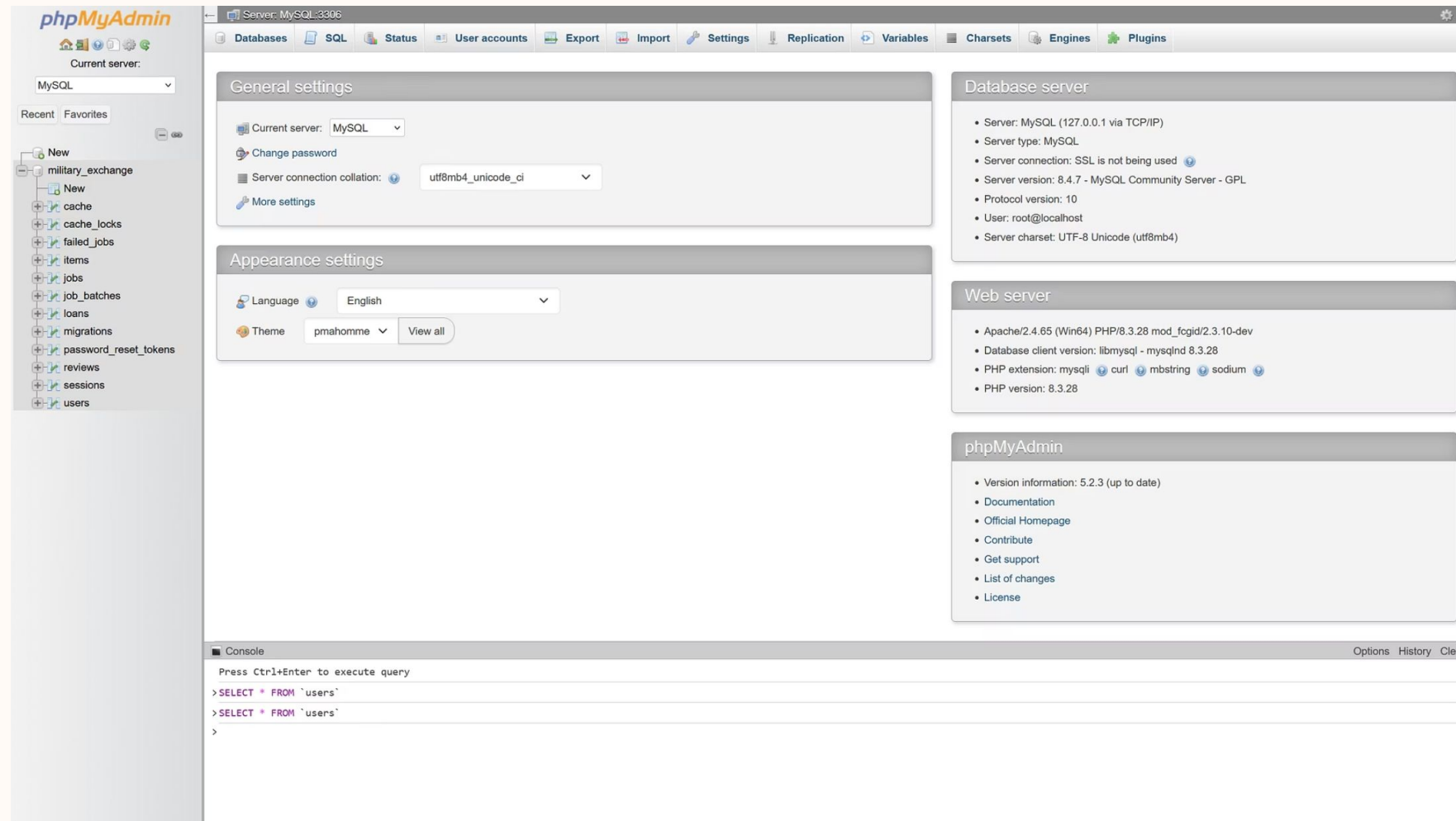
Managed via **WAMP** with MySQL persistence, enabling rapid local development and testing before production deployment.



Wampserver

Apache 2.4 - MySQL 5, 8 & 9 - MariaDB 10 & 11 - PHP 5, 7 & 8

Data Integrity: The Chain of Custody



Relational Schema

A structured **MySQL database** powered by Eloquent ORM ensures typed, queryable records for every piece of equipment in the system.

Many-to-Many Relationships

A dedicated **loans pivot table** tracks item possession over time, creating an auditable chain of custody for every transaction.

Real-Time Persistence

Automated status updates ensure gear is never "lost" in the system — every item reflects its current state: **Secured** or **Lent**.

PRIVATE INTELLIGENCE

Local Ollama Integration

8-Billion Parameter LLM

Powered by **Llama3** running locally via Ollama — no cloud dependency, no external API costs.

100% Private Processing

A **4.7GB local footprint** keeps all logistical data on-machine. No data ever leaves the local environment.

Guzzle HTTP Bridge

Laravel communicates with the Ollama service via **Guzzle**, automating item categorization and review sentiment analysis.



Power OpenClaw with Ollama

The easiest way to build
with open models

```
irm https://ollama.com/install.ps1 | iex
```

paste this in PowerShell, or [download Ollama](#)

Integrity First: The Reputation Engine

```

1 | time=2026-04-15T13:25:21.435-07:00 level=INFO source=routes.go:1752 msg="server config" env="map[CUDA_VISIBLE_DEVICES: GGML_VK_VISIBLE_DEVICES: GPU_DEVICE_ID: 0]"
2 | time=2026-04-15T13:25:21.437-07:00 level=INFO source=routes.go:1754 msg="Ollama cloud disabled: false"
3 | time=2026-04-15T13:25:21.446-07:00 level=INFO source=images.go:499 msg="total blobs: 5"
4 | time=2026-04-15T13:25:21.447-07:00 level=INFO source=images.go:506 msg="total unused blobs removed: 0"
5 | time=2026-04-15T13:25:21.474-07:00 level=INFO source=routes.go:1810 msg="Listening on 127.0.0.1:11434 (version 0.20.7)"
6 | time=2026-04-15T13:25:21.479-07:00 level=INFO source=runner.go:67 msg="discovering available GPUs..."
7 | time=2026-04-15T13:25:21.490-07:00 level=INFO source=runner.go:106 msg="experimental Vulkan support disabled. To enable, set OLLAMA_VULKAN=1"
8 | time=2026-04-15T13:25:21.505-07:00 level=INFO source=server.go:444 msg="starting runner" cmd="C:\\Users\\68107\\AppData\\Local\\Programs\\Ollama\\ollama.exe"
9 | time=2026-04-15T13:25:35.897-07:00 level=INFO source=server.go:444 msg="starting runner" cmd="C:\\Users\\68107\\AppData\\Local\\Programs\\Ollama\\ollama.exe"
10 | time=2026-04-15T13:25:36.679-07:00 level=INFO source=server.go:444 msg="starting runner" cmd="C:\\Users\\68107\\AppData\\Local\\Programs\\Ollama\\ollama.exe"
11 | time=2026-04-15T13:25:37.096-07:00 level=INFO source=types.go:60 msg="inference compute" id=cpu library=cpu compute="" name=cpu description=cpu libdirs=ollama
12 | time=2026-04-15T13:25:37.096-07:00 level=INFO source=routes.go:1860 msg="vram-based default context" total_vram="0 B" default_num_ctx=4096
13 | [GIN] 2026/04/15 - 13:25:37 | 200 | 3.1482ms | 127.0.0.1 | GET "/api/version"
14 | time=2026-04-15T16:27:16.276-07:00 level=INFO source=cpu_windows.go:148 msg="packages count=1"
15 | time=2026-04-15T16:27:16.286-07:00 level=INFO source=cpu_windows.go:195 msg="" package=0 cores=4 efficiency=0 threads=8
16 | llama_model_loader: loaded meta data with 22 key-value pairs and 291 tensors from C:\\Users\\68107\\ollama\\models\\blobs\\sha256-6a0746a1ec1aef3e7ec53868f220f
17 | llama_model_loader: Dumping metadata keys/values. Note: KV overrides do not apply in this output.
18 | llama_model_loader: - kv 0: general.architecture str = llama
19 | llama_model_loader: - kv 1: general.name str = Meta-Llama-3-8B-Instruct
20 | llama_model_loader: - kv 2: llama.block_count u32 = 32
21 | llama_model_loader: - kv 3: llama.context_length u32 = 8192
22 | llama_model_loader: - kv 4: llama.embedding_length u32 = 4096
23 | llama_model_loader: - kv 5: llama.feed_forward_length u32 = 14336
24 | llama_model_loader: - kv 6: llama.attention.head_count u32 = 32
25 | llama_model_loader: - kv 7: llama.attention.head_count_kv u32 = 8
26 | llama_model_loader: - kv 8: llama.rope.freq_base f32 = 500000.000000
27 | llama_model_loader: - kv 9: llama.attention.layer_norm_rms_epsilon f32 = 0.000010
28 | llama_model_loader: - kv 10: general.file_type u32 = 2
29 | llama_model_loader: - kv 11: llama.vocab_size u32 = 128256
30 | llama_model_loader: - kv 12: llama.rope.dimension_count u32 = 128
31 | llama_model_loader: - kv 13: tokenizer.ggml.model str = gpt2
32 | llama_model_loader: - kv 14: tokenizer.ggml.pre str = llama-bpe
33 | llama_model_loader: - kv 15: tokenizer.ggml.tokens arr[str,128256] = ["!", "\"", "#", "$", "%", "&", "'", ...
34 | llama_model_loader: - kv 16: tokenizer.ggml.token_type arr[i32,128256] = [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
35 | llama_model_loader: - kv 17: tokenizer.ggml.merges arr[str,280147] = ["ĠĠ", "ĠĠĠ", "ĠĠĠĠ", "...
36 | llama_model_loader: - kv 18: tokenizer.ggml.bos_token_id u32 = 128000
37 | llama_model_loader: - kv 19: tokenizer.ggml.eos_token_id u32 = 128009
38 | llama_model_loader: - kv 20: tokenizer.chat_template str = {% set loop_messages = messages %}{% ...
39 | llama_model_loader: - kv 21: general.quantization_version u32 = 2
40 | llama_model_loader: - type f32: 65 tensors
41 | llama_model_loader: - type q4_0: 225 tensors

```

Above is the **Ollama server log** showing the Llama-3-8B model loading into memory.

AI Sentiment Analysis

Rather than simple star ratings, the system uses **AI sentiment analysis** to evaluate the full text of peer reviews — detecting nuance that numeric scores miss.

Dynamic Reliability Rating

User scores update automatically based on peer feedback. A positive review raises the **Reliability Rating**; negative sentiment lowers it.

Enforced Accountability

Reputation scores directly impact borrowing privileges. Low scores trigger **restriction gates**, preventing unreliable users from accessing community resources.

Role-Based Access Control (RBAC)

Security is modeled on military hierarchy. Custom **Laravel Middleware** enforces permissions by rank, ensuring appropriate access at every level of command.

BORROW GEAR

SUBMIT REVIEW / APPEAL

Target Email:

Rating (1-10):

Detailed Review/Complaint:

SUBMIT ANALYSIS

Hierarchy-Driven Permissions

Custom middleware intercepts every request and validates the user's rank before granting access to administrative functions.

Authentication

Secured via **Laravel Breeze** with encrypted sessions and protected routing.

 CURRENTLY CHECKED OUT BY:

E-1 Evyn Baker

60%

Resource & State Management

AVAILABLE EQUIPMENT

Total Resources: 19

Sledgehammer

GENERAL | CONDITION: GOOD

a heavy metal stick for hitting things

Owner Score: 60%

CURRENTLY CHECKED OUT BY:

E-1 Evyn Baker

60%

BORROW GEAR

5x8 Utility Trailer

CLASS VII | CONDITION: GOOD

Reliable, well-maintained equipment for base community use.

Owner Score: 60%

BORROW GEAR

Dewalt 20V Max Cordless Drill

CLASS IX | CONDITION: GOOD

Reliable, well-maintained equipment for base community use.

Owner Score: 60%

CURRENTLY CHECKED OUT BY:

E-1 Evyn Baker

60%

BORROW GEAR

Tactical Rucksack (OCP)

CLASS II | CONDITION: GOOD

Reliable, well-maintained equipment for base community use.

Owner Score: 60%

CURRENTLY CHECKED OUT BY:

E-1 Evyn Baker

60%

BORROW GEAR

Heavy Duty Floor Jack

CLASS IX | CONDITION: GOOD

Reliable, well-maintained equipment for base community use.

Owner Score: 60%

CURRENTLY CHECKED OUT BY:

Unknown User

7%

BORROW GEAR

KitchenAid Stand Mixer

GENERAL | CONDITION: GOOD

Reliable, well-maintained equipment for base community use.

Owner Score: 60%

BORROW GEAR

Portable Pressure Washer

CLASS VII | CONDITION: GOOD

Reliable, well-maintained equipment for base community use.

Owner Score: 60%

CURRENTLY CHECKED OUT BY:

E-1 Evyn Baker

60%

BORROW GEAR

Camping Tent (4-Person)

CLASS II | CONDITION: GOOD

Reliable, well-maintained equipment for base community use.

Owner Score: 60%

BORROW GEAR

RELIABILITY RATING

15

DESIGNATION

O-5 Sawyer Wilson

Rank: O-5

SYNC RELIABILITY NODE (AI)

Recalculates status based on latest peer reviews using Ollama Sentiment Engine.

POSSESSIONS TRACKING

EQUIPMENT	STATUS	DUE DATE	ACTIONS
BIG Hammer	SECURED	Apr 29, 2026	RETURNRENEW
biggerer wrench damaged	SECURED	Apr 29, 2026	RETURNRENEW

Diverse Inventory

Tracks **17 resource types (or more)** — from heavy equipment like Utility Trailers to tactical gear like Rucksacks — with full classification and status history.

Enforced Borrowing Logic

A hard **5-item borrow limit** per user ensures gear remains available across the community. Automated due-date management prevents indefinite possession cycles.

State Tracking

Real-time **"Secured"** vs **"Lent"** status with automated reminders and renewal logic keeps the inventory accurate at all times.

MISSION

STATUS

Technical Challenges & Future Growth

Hurdles Overcome

Successfully resolved **SQL Error 1054** during schema alignment and managed PHP versioning conflicts. High-latency local AI processing was mitigated through strategic user feedback loops.

Future Vision

Planned **QR code integration** for instantaneous gear hand-offs in the field. A "Simple Mode" toggle will offer users an option to bypass local AI analysis for faster interactions.

Final Status

Mission Ready. The MRE delivers a secure, intelligent, and accountable equipment exchange platform — purpose-built for the Air Force community.



Thanks

Evyn w. Baker